MODELING NATURAL GAS COMBUSTION DYNAMICS

By David Wagman

An ongoing challenge for natural gas turbine manufacturers is to simultaneously achieve low NO_x emissions and flame stability. The balancing act depends on running a lean fuel mixture during combustion. Too rich a mixture and the unit risks exceeding air quality compliance standards. Too lean a mixture and the turbine can suffer loss of ignition.

"Gas turbines run near the edge of lean blowoff where the flame is always on the hairy edge," said Bernie Rosenthal, CEO of Reaction Design. The California-based company, founded in 1997 by two MIT professors, markets software that helps turbine designers understand the chemical reactions that occur when natural gas is burned in a combustion turbine. Some 350 commercial customers are currently using Reaction Design's Chemkin product, whose original use was to measure rocket engine combustion chemistry.

Chemkin is based in part on a fuel chemistry database, which consists of fundamental chemical descriptions. By altering variables such as catalysts, temperature and pressure, the model can predict not only what types of emissions are formed when the natural gas burns, but where in the combustion process emissions will emerge and when.

Natural gas is a fairly basic molecule, Rosenthal said, and the prediction can be fairly straightforward. The models grow in complexity, however, as synthetic and alternative natural gas sources are burned.

"As you add combinations of fuels, the effect becomes less well understood," he said. Mixing natural gas with syngas can create effects not seen before. Those unseen effects drive an interest to understand what is happening during combustion.

Rosenthal said he sees Chemkin as a tool to fill a knowledge gap that exists for turbine designers and engineers.

"One potential game-changing innovation is the idea that you can map your geometry and CFD (computational fluid dynamics) results to an equivalent reactor network," he said. Chemists use different reactor models as a way to understand what is happening in the combustion chamber. The models allow for detailed snapshots of mixing, flame, recirculation and post-flame segments within the chamber.

"You can start to see when emissions are created and flame stability with regard to regions where certain chemical reactions take place," he said.

The chemical analysis has potential application in pulverized coal combustion boilers as well as in gas turbines. In the case of coal, unburned hydrocarbons and particulates are primary issues. With carbon regulation seen as almost inevitable, the ability to predict how much carbon dioxide will be produced through combustion is important.

Rosenthal said many basic fuels are already well understood, but only to the extent that we know how natural gas is being burned. "There's a myth that we really understand what's going on at the chemical level" in gas turbine combustors. A great deal of empirical understanding exists when it comes to combustion chemistry, but relatively little scientifically-based knowledge is available. With new fuels (such as syngas and landfill gas) experimental data can be gathered on an ever-widening universe of fuel options, but typically at the expense of time and money.

"There is a real interest on the part of turbine manufacturers to understand those dynamics with computer modeling," Rosenthal said. Software such as that developed and marketed by Reaction Design may help accelerate the process of understanding the implications of combusting fuels in turbines.

Capstone MicroTurbine™

Your Green and Clean Building Energy Solution

- Provides Building Heating, Cooling or Hot Water
- Provides Continuous Electric Power With or Without the Electric Utility
- High Efficiency System lowers Greenhouse Gas Emissions
- Air Bearings Eliminate Liquid Lubricants & Disposal of Hazardous Materials
- Extremely Low NOx Emissions
- Qualifies for LEED Points for LEED Certification
- Reduces Your Energy Costs and Increases Power Security
- Thousands of Units Shipped Worldwide
- Millions of Documented Runtime Operating Hours
- Operates Unattended and Lowers Maintenance Costs
- Backed by Capstone Factory Service
- Meets Local Building Codes, Utility Interconnection & Air-Permitting Requirements

Power When and Where You Need it. Clean and Simple.

Call Now To Learn How Capstone MicroTurbines™ Can Help You

866.422.7786 • www.microturbine.com

Capstone

For info. http://powereng.hotims.com RS# 29

Copyright of Power Engineering is the property of Penn Well Publishing Co. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.